

# Advanced Mathematical Techniques (AMT)

## Notes

### 1 Today's stuff

Let's try and solve:

$$\int_0^\infty \frac{\ln u \ln(u+1)}{(u+1)^2} du$$

and:

$$\int_0^\infty \frac{\ln u \ln(u+1)}{u+1} du$$

Know that if the only issue with an expression is that the logarithm goes to infinity, that's not an issue. Consider  $\ln x$  vs  $e^{-x}$ . We see that  $e^{-x}$  always takes precedence because it decreases so fast, so we can think of it as the big winner, in that whatever it wants to happen will happen. Similarly, the  $\ln$  is the big loser, because the logarithm isn't good enough to make things converge or diverge.

We call the divergence that the logarithm produces **integrable**.

$$u \rightarrow 0, \ln u \rightarrow -\infty$$

This is ok because the logarithm is weak and its divergence is **integrable**. However, as  $u \rightarrow \infty$ , the integral behaves like:

$$\int_0^\infty \frac{\ln^2 u}{u}$$

Let's sidestep this and do some truncated BC calculations.

$$\int_c^\infty \frac{du}{u^p} = \frac{u^{-p+1}}{-p+1} \Big|_c^\infty$$

Clearly, this is okay if  $p > 1$ . Also, as  $u$  goes to infinity, dividing by  $u$  helps things converge (turns 1 into harmonic), though it doesn't work fully. Therefore, dividing by 2 powers of  $u$  is even better!

That's the behavior around  $\infty$ , so let's find the behavior around 0. Also,

$$\int_0 \frac{du}{u^p}$$

As  $u$  goes to 0, dividing by  $u$  hurts convergence, so we want to divide by less than 1 whole power. Notice that  $p = 1$  gives nothing helpful - it would give a logarithm after the integration, which would be undefined at 0 and  $\infty$ .

Going back to our problem, we know that the logarithms are weak and we only have one power of  $u$  in the denominator, so we're good.

$$\int_0^\infty \frac{\ln u \ln(u+1)}{u+1} du \text{ diverges.}$$

Also, we know that since the first integral has a power of 2 in the denominator, it converges.

$$\int_0^\infty \frac{\ln u \ln(u+1)}{(u+1)^2} du$$

Consider instead:

$$\int_0^\infty \frac{u^\varepsilon (u+1)^\delta}{(u+1)^2} du$$

We want the coefficient of  $\varepsilon\delta$ . Rigorously (at least partially),

$$\int_0^\infty \frac{u^\varepsilon (u+1)^\delta}{(u+1)^2} du = \int_0^\infty \frac{e^{\varepsilon \ln u} e^{\delta \ln(u+1)}}{((u+1)^2)} du$$

For small  $\varepsilon, \delta$  (which is assumed by the variable names themselves):

$$\sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \frac{\varepsilon^n \delta^m}{n!m!} \int_0^{\infty} \frac{\ln^n u \ln^m(u+1)}{(u+1)^2} du$$

Solving by  $B_2$ ,

$$\begin{aligned} \int_0^{\infty} \frac{u^{\varepsilon}}{(u+1)^{2-\delta}} du &= \frac{\Gamma(1+\varepsilon)\Gamma(1-\varepsilon-\delta)}{\Gamma(2-\delta)} = \frac{\Gamma(1+\varepsilon)\Gamma(1-\varepsilon-\delta)}{(1-\delta)\Gamma(1-\delta)} \\ &= \frac{1}{1-\delta} \exp\left(-\gamma\varepsilon + \sum_{k=2}^{\infty} \frac{(-1)^k \zeta(k)}{k} \varepsilon^k - \gamma(-\varepsilon-\delta) + \sum_{k=2}^{\infty} \frac{(-1)^k \zeta(k)}{k} (-\varepsilon-\delta)^k - \left[\gamma\delta + \sum_{k=2}^{\infty} \frac{(-1)^k \zeta(k)}{k} (-\delta)^k\right]\right) \end{aligned}$$

The only contribution to the  $\varepsilon\delta$  is that one binomial term. Thus, the coefficient being 2 and the sum's term being 2 cancel to give us that the final answer is  $\zeta(2)$ . Similar logic shows that by expanding in terms of the zeta function gives us the final answer for the second term being squared in the integral is  $2(\zeta(2) + \zeta(3))$ .

On to  $B_3$ ,

$$\int_0^{\infty} x^{\alpha-1} e^{-x} dx \cdot \int_0^{\infty} y^{\beta-1} e^{-y} dy = \int_0^{\infty} dx \int_0^{\infty} dy x^{\alpha-1} y^{\beta-1} e^{-x-y}$$

Let  $x = u^2, y = v^2$ ,

$$\Gamma(\alpha)\Gamma(\beta) = \int_0^{\infty} 2udu \int_0^{\infty} 2vdv \cdot u^{2\alpha-2} v^{2\beta-2} e^{-u^2-v^2}$$

By polar,

$$\begin{aligned} &= 4 \int_0^{\infty} r dr \int_0^{\frac{\pi}{2}} d\theta (r \cos \theta)^{2\alpha-1} \cdot (r \sin \theta)^{2\beta-1} e^{-r^2} \\ &= 4 \int_0^{\infty} r^{2\alpha+2\beta-1} e^{-r^2} \cdot \int_0^{\frac{\pi}{2}} \cos^{2\alpha-1} \theta \sin^{2\beta-1} \theta d\theta \end{aligned}$$

Let  $r^2 = t$ .

$$= 2 \int_0^{\infty} t^{\alpha+\beta-1} e^{-t} \cdot \int_0^{\frac{\pi}{2}} \cos^{2\alpha-1} \theta \sin^{2\beta-1} \theta d\theta$$

From the definition of  $B_1$ :

$$\int_0^{\frac{\pi}{2}} \cos^{2\alpha-1} \theta \sin^{2\beta-1} \theta d\theta = B_3 = \frac{\Gamma(\alpha)\Gamma(\beta)}{2\Gamma(\alpha+\beta)}$$

Let's test with:

$$\frac{0}{\frac{\pi}{2}} \cos^4 \theta \sin^6 \theta d\theta = \frac{\Gamma\left(\frac{5}{2}\right) \cdot \Gamma\left(\frac{7}{2}\right)}{2\Gamma(6)} = \boxed{\frac{3\pi}{512}}$$